Fragment-Based Invariant Subgraph Extraction for Out-of-Distribution Generalization on Molecular Graphs

Nancy Yang
Department of Computer Science
Northwestern University
Evanston, IL, USA
nancyyang500@gmail.com

Lance Santiago
Department of Computer Science
Georgia State University
Atlanta, GA, USA
lsantiago4@student.gsu.edu

***Abstract*—Graph neural networks (GNNs) have achieved remarkable success in molecular property prediction but often fail to generalize under distribution shifts, where spurious correlations learned during training do not transfer well to new distributions beyond the training data. Invariant subgraph learning offers a way to improve robustness and interpretability by identifying predictive subgraphs that remain reliable across distributions. This is particularly important in molecular property prediction, where properties depend on chemical substructures and the structural relationships among them. Many existing invariant-subgraph approaches operate on atom-level graphs, which can produce rationales that are difficult to interpret chemically or capture incomplete subsets of meaningful molecular motifs. To address this, we propose a fragment-based invariant subgraph learning approach that uses fragment-level molecular representations as a more chemically structured and interpretable space for stable predictive evidence. Our framework decomposes molecules into fragment graphs whose nodes represent connected molecular substructures, selects and aggregates rationales across multiple GNN views, and reconstructs this evidence into atom-level subgraphs to preserve chemical detail during graph classification. We evaluate how this approach affects out-of-distribution (OOD) molecular property prediction and analyze selected rationales to identify chemically meaningful molecular motifs, providing a more interpretable view of model predictions.**

***Index Terms*—molecular property prediction, graph neural networks, out-of-distribution generalization, invariant subgraph learning, molecular fragmentation, interpretability**

I. Introduction

Molecular property prediction is a natural application of graph neural networks (GNNs). A molecule can be represented as a graph where atoms are nodes and chemical bonds are edges, allowing a model to learn from both atomic properties and molecular connectivity. This representation has made GNNs useful for tasks that support drug discovery, including property prediction.

Strong performance on molecules drawn from the same distribution as the training data is not enough, however. A model can learn correlations that are useful in its training data but unreliable for molecules from shifted distributions, such as those arising from different assays, scaffold families, or molecular sizes. The out-of-distribution (OOD) problem is particularly important in molecular discovery, where predictive models are expected to generalize beyond the chemical space represented in the training set. Correlation alone does not establish a stable chemical relationship.

Invariant-subgraph learning is one approach to this problem. Rather than relying on every part of a graph, an invariant-subgraph method seeks a smaller predictive subgraph, often called a rationale, whose relationship to the label is more stable across environments. This goal is especially important for molecules where a label may depend on particular chemically meaningful substructures like functional groups, ring systems, or relationships among substructures. Yet most graph-rationale methods select evidence directly from atom-level edges. Atom-level selection preserves fine structural detail, but the resulting rationale can divide a recognizable chemical motif across selected and unselected bonds. Such a rationale may be useful to the model while remaining difficult for a chemist to inspect and analyze.

We therefore investigate a fragment-first design. Each molecule is represented in two linked forms. The atom graph retains the original atoms and bonds. A fragment graph groups connected atoms into molecular substructures and records the bonds and attachment atoms that connect those fragments. The model selects relationships in fragment space, then maps the selected evidence back into an atom-level subgraph before making the molecular prediction. This gives the selection mechanism chemically structured candidate units without discarding the atom-level detail needed for classification.

Our approach is motivated by the idea that multiple invariant subgraphs can provide complementary evidence. Multiple views may capture complementary evidence, rather than forcing a single rationale to explain an entire molecule. In our framework, views using the same fragmentation method aggregate compatible fragment-edge scores. A molecule may contain several candidate predictive structures rather than a single informative motif. We therefore use multiple model views to identify complementary fragment relationships and combine their predictions.

Our contributions are:

- We introduce a fragment-first invariant subgraph learning framework for molecular OOD prediction. The framework selects relationships between molecular fragments, reconstructs the selected evidence into an atom-level rationale, and performs graph classification while retaining atom-and-bond chemical detail.
- We extend multiple invariant subgraph extraction to molecular fragment graphs, supporting Group Graph [1], BRICS [2], RECAP [3], and mixed fragmentation configurations. Compatible views aggregate fragment-edge evidence, while different fragmentation-method groups combine predictions through soft voting.
- We provide a workflow for chemical interpretability that analyzes reconstructed atom-level rationales, the selected fragment, connected fragment pairs, and agreement across model views. These analyses identify model-relevant candidate molecular evidence.

II. Related Work

A. Molecular Graph Fragmentation Representations

Although atom-level graph neural networks and invariant-subgraph methods can achieve strong predictive performance, the rationales they select may be difficult to interpret in chemical terms. An atom-level rationale can divide a recognizable molecular motif across selected and unselected bonds, making it harder to summarize the model evidence as a coherent substructure.

Molecular fragmentation partitions a molecule into connected substructures that can serve as higher-level units for representation and analysis. Many chemically motivated fragmentation methods use predefined structural patterns or bond-cleavage rules to create these partitions. Group Graph represents molecules through non-overlapping substructures derived from predefined functional-group patterns, aromatic rings, and fatty carbon groups, while retaining attachment information at fragment boundaries. BRICS and RECAP instead apply rule-based bond-cleavage strategies motivated by chemically plausible molecular decompositions. Because these methods partition the same molecule differently, they provide alternative fragment-level views of possible molecular evidence.

Fragment representations do not establish that a selected fragment is chemically causal for a molecular property. Instead, they provide chemically structured candidate units that can make model-selected evidence easier to inspect, compare across molecules, and prioritize for further chemical investigation.

B. Graph OOD & Invariant Subgraph Learning

Out-of-distribution generalization seeks models whose predictions remain reliable when test samples differ from the training distribution. Invariant learning methods aim to identify predictive evidence that remains useful across environments instead of relying on environment-specific correlations. CIGA [4] frames this through a causal graph-generation model: a featurizer selects an estimated invariant subgraph, and a classifier predicts the label from that selection. Its objective combines label prediction with same-label contrastive learning, encouraging selected subgraphs from different graphs with the same class to retain shared class-related information while excluding spurious graph structure or attributes. Among graphs with the same label, the causal evidence more likely recurs even when background structure, attributes, scaffold or assay context, and other environmental factors change. A shortcut should be less consistently shared.

This is appealing because molecular properties may depend on particular functional groups, ring systems, and relationships between substructures rather than every atom in a molecule. However, CIGA and many invariant-subgraph approaches perform selection directly on atom-level graphs. These selections can preserve fine detail but may divide recognizable chemical motifs across the rationales, limiting their readability as molecular evidence. Our work studies invariant-subgraph learning in a fragment graph, where the selectable units are chemically structured candidate subgraphs. The selected fragment-level evidence is then reconstructed as an atom-level rationale for downstream molecular graph prediction.

C. Subgraph Aggregation

A single invariant subgraph may not capture all predictive evidence in a graph. We follow the methodology of SuGAr [5], which addresses this limitation by learning multiple diverse invariant subgraphs and aggregating their information for OOD prediction. Its central motivation is that multiple predictive structures may exist and that combining complementary views can produce a more robust prediction than relying on one selected subgraph.

Our framework adapts this multi-view principle to fragment-level molecular invariant learning. Separate FragmentCIGA views score fragment relationships and produce fragment-edge rationales; these are reconstructed into atom-level subgraphs before classification. Within a fragmentation method, compatible fragment-edge scores can be aggregated before reconstruction. Across different fragmentation methods, whose partitions are not directly aligned, we combine prediction probabilities through soft voting. Thus, the method combines fragment-level rationale selection, atom-level prediction, and inference-time aggregation

III. Methodology

We represent each molecule at two connected levels. The atom graph preserves the complete molecular structure used for prediction, while a fragmentation procedure defines a higher-level fragment graph on which rationale selection is performed. The selected fragment-level evidence is then mapped back to the original atom graph. This lets the model select chemically structured candidate units while retaining atom-level detail for classification.

A. Preprocessing

1) Atom Graph Construction & Storage: Each molecular SMILES string is parsed and sanitized with RDKit. We construct an atom graph in which atoms are nodes and chemical

bonds are edges. Each record retains atom features, bond features, the classification label, a molecule identifier, and available dataset metadata, including domain or environment identifiers. Domain metadata is retained for provenance and optional training objectives; it is not added to the molecular node or edge features used by the classifier.

For each molecule, preprocessing stores a canonical undirected bond table and corresponding directed edges for message passing. The canonical table preserves original atom and bond indices, which are needed to reconstruct selected rationales later in the pipeline. Processed examples are stored as split-level PyTorch Geometric files together with metadata describing the dataset, fragmentation setup, feature-pooling strategy, and preprocessing version.

2) Fragmentation: Each fragmentation method produces a non-overlapping partition of the original atom graph. A fragmentation result contains connected fragments, an atom-to-fragment mapping, fragment-boundary edges, and provenance mappings from each fragment edge to the original molecular bond and its two attachment atoms. These mappings ensure that every atom is assigned to exactly one fragment and that selected fragment-level evidence can be reconstructed in the original molecular graph. The framework supports Group Graph, BRICS, and RECAP fragmentation. In mixed-fragmentation configurations, these methods define separate fragment partitions over the same underlying atom graph rather than altering the atom graph itself.

Group Graph provides chemically motivated candidate clusters, including functional groups, aromatic structures, and carbon-chain-related groups. Because candidate clusters can overlap, we convert them into a deterministic molecule-wide partition suitable for feature pooling and atom-level reconstruction. The overlap policy prioritizes split fragments, then larger candidate clusters, followed by stable atom-index tie-breaking. A higher-priority cluster claims shared atoms first; lower-priority clusters retain only unassigned atoms. If a residual cluster becomes disconnected after overlap resolution, it is divided into connected components. Any atom not assigned by the upstream clustering rules is included in a fallback fragment. This produces connected, non-overlapping fragments with complete atom coverage and stable provenance mappings.

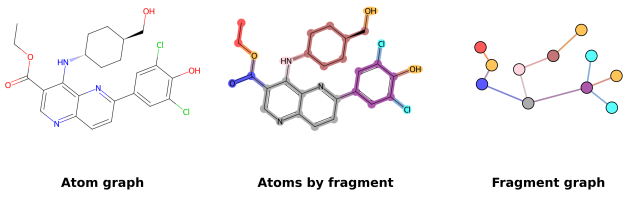


Fig. 1. Example fragment graph created by Group Graph.

BRICS and RECAP provide rule-based alternatives to the Group Graph partition. BRICS identifies RDKit BRICS-cleavable bonds, removes those bonds, and treats the connected components of the remaining original atom graph as fragments.

RECAP is implemented by projecting matched RECAP cleavage rules onto bonds in the original molecule and then computing connected components after removing the matched boundary bonds. This projection preserves every original atom, including atoms that could otherwise be omitted or represented as dummy atoms in a product-side RECAP decomposition. Thus, BRICS and RECAP provide alternative fragment-level views of the same molecule while preserving the provenance needed for feature pooling, rationale reconstruction, and comparison across views.

3) Fragment Graph Construction and Feature Pooling: For each fragmentation, we construct a fragment graph whose nodes represent fragments and whose edges represent original bonds crossing fragment boundaries. Fragment-node features are computed from the underlying atom graph.

To reduce the dimensionality of the atom representation before pooling, the original atomic-number encoding is replaced with a compact encoding for 13 common drug-like elements: H, B, C, N, O, F, Si, P, S, Cl, Se, Br, I, and an unknown category. Degree, formal charge, chirality, hydrogen count, hybridization, aromaticity, and scaled atomic mass are retained. This yields a 46-dimensional atom feature vector. For each fragment, we mean-pool features from its assigned atoms and from bonds whose endpoints both lie within that fragment. We then append the fragment's atom count and internal-bond count. The resulting static fragment feature has 62 dimensions: 46 atom features, 14 bond features, and 2 size features. Fragment-edge attributes concatenate the feature vectors of the source and destination attachment atoms, the feature vector of the original boundary bond, and the sizes of the two endpoint fragments. These attributes preserve information about how fragments are connected in the original molecule.

B. Training

1) Fragment-Edge Sampling and Rationale Selection: Each FragmentCIGA view operates on a fragment graph whose nodes are molecular fragments and whose edges correspond to original bonds that cross fragment boundaries. Before scoring, each view independently applies training-time random fragment-edge sampling. For a configured keep ratio, the sampler marks a subset of fragment edges as eligible for that forward pass while preserving all fragment nodes and the original identity of every retained edge. The sampled graph is therefore a stochastic view of the same molecular fragmentation rather than a newly fragmented molecule.

Random edge sampling serves two purposes. First, it prevents a view from depending on one fixed fragment-graph connectivity pattern during training. Second, because views use independent sampling seeds, it gives otherwise identically initialized views different optimization trajectories and helps the multi-view system discover complementary rationale patterns. Sampling is active only during training; at inference, each view scores the full fragment graph.

After sampling, a fragment-level GNN produces a continuous importance score for every eligible fragment edge. Scores from the sampled graph are mapped back to the original fragment-edge ordering so that they remain aligned with the stored reconstruction provenance. For each molecule, the model selects the highest-scoring eligible fraction of fragment edges, determined by an edge ratio. This hard top-k mask defines the discrete rationale used to determine which fragments and boundary bonds are reconstructed.

The model also retains continuous edge scores as soft importance values. Thus, the hard top-k operation determines the atom-level subgraph included in the rationale, while the continuous scores modulate message passing in the reconstructed atom graph. This combination makes the rationale interpretable as a discrete set of fragment relationships while allowing classification loss to provide gradient-based learning signals to the fragment-edge scorer.

2) Atom Reconstruction and Classification: A selected fragment edge identifies its two endpoint fragments. Using stored atom-to-fragment assignments and fragment-edge provenance, the model reconstructs an atom-level rationale containing the atoms of selected fragments, bonds internal to those fragments, and original boundary bonds corresponding to selected fragment edges.

This reconstruction separates the representation used for selection from the representation used for prediction. Fragment graphs provide chemically structured units for rationale selection, while the reconstructed atom graph retains detailed atom and bond information for molecular classification.

A GIN-style atom-level classifier processes each reconstructed rationale. Fragment-edge importance scores are retained as message-level edge gates in the reconstructed atom graph, allowing atom-level message passing to reflect fragment-level importance. Consequently, prediction loss can backpropagate through soft importance scores to train the fragment-edge scorer.

3) Training Objectives: The core objective is prediction loss: each view is trained to predict the molecular label from its reconstructed atom-level rationale. The training pipeline also supports configurable auxiliary objectives. Supervised contrastive loss encourages graph-level representations with the same label to be similar and representations with different labels to be distinct. Diversity loss penalizes overly similar fragment-edge score patterns among compatible views, encouraging different views to attend to different fragment relationships. The optional IRM penalty discourages variation in predictive behavior across training environment groups when meaningful environment data is available.

For a set of views, the total training loss combines these

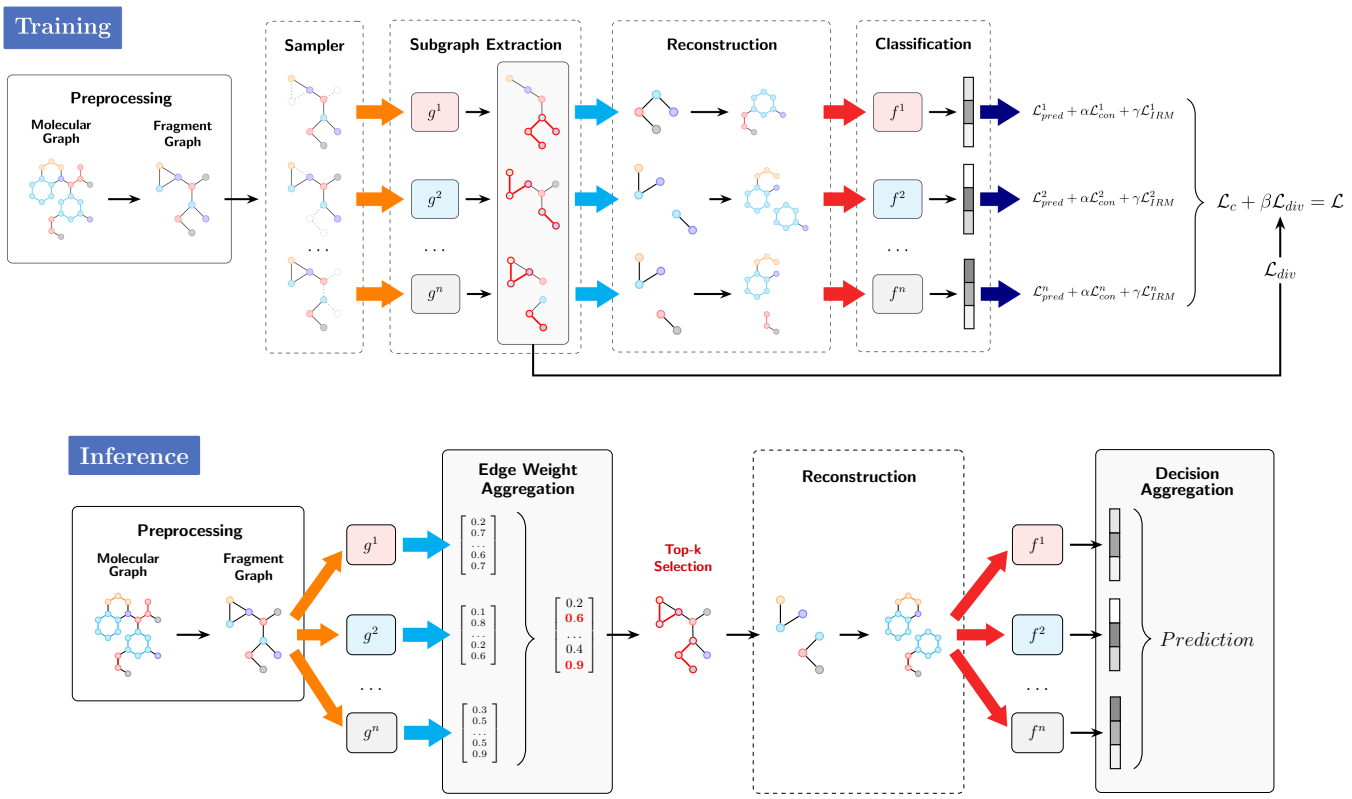


Fig. 2. **Visualization of our model's training and inference pipelines.** (a) During training, the sampler randomly drops edges from the fragment graph for each view. Featurizers g^i predict edge weights, top-k edges form the extracted subgraph, which is reconstructed into an atom-level subgraph for classification alongside diversity loss $\mathcal{L}_{div}$. (b) During inference, featurizer edge scores are aggregated, the top-k edges form the aggregated subgraph, which is reconstructed into an atom-level subgraph to yield final soft-vote predictions across classifiers f^i.

terms as:

$$L = L_{\text{pred}} + \lambda_{\text{con}} L_{\text{con}} + \lambda_{\text{div}} L_{\text{div}} + \lambda_{\text{irm}} L_{\text{irm}}$$

The prediction term is the sum of negative log-likelihood losses across views. It ensures that each view's reconstructed atom-level rationale remains sufficient for molecular classification. Because the classifier receives importance-gated atom-level messages, this term also trains fragment-edge scoring toward predictive fragment relationships.

The supervised contrastive term is computed from graph-level embeddings produced by the atom-level classifiers. Embeddings from examples with the same class label are treated as positive pairs, while examples with different labels act as negatives. This encourages each view to organize selected-rationale representations by class rather than solely by molecule-specific features.

The diversity term is applied only among views that use the same fragmentation method, since their fragment-edge vectors are aligned. It measures the similarity of per-molecule edge-score patterns between pairs of views and penalizes positive similarity. Minimizing this term encourages views to assign importance to different fragment relationships instead of collapsing onto the same rationale. This is the mechanism that gives the multi-view ensemble a reason to provide complementary evidence rather than several copies of the same selector.

The optional IRM term uses training environment assignments to penalize variation in predictive behavior across environment groups. When enabled, examples sharing an environment key are assigned to the same group, and the model is encouraged to maintain a more stable predictor across those groups. This term should be interpreted as environment-based regularization when the chosen grouping corresponds to a meaningful source of distribution shift.

C. Inference

1) Rationale Selection: For an unseen molecule, each trained view scores fragment edges without updating model parameters. Views using the same fragmentation method operate over aligned fragment-edge identities and can therefore aggregate their scores using the configured mean or maximum rule. The aggregated scores are used to select the highest-scoring fragment edges for that fragmentation method.

When multiple fragmentation methods are enabled, each method retains its own fragment graph and produces its own method-specific rationale. Scores are not averaged across different fragmentation schemes because their fragment nodes and edges do not necessarily correspond directly.

2) Reconstruction and Prediction: For each fragmentation method, selected fragment edges are reconstructed into an atom-level rationale using the same provenance mappings used during training. Each retained view classifier then produces class probabilities from the reconstructed rationale associated with its fragmentation method.

3) Final Soft-vote Prediction: The final prediction is obtained by averaging class probabilities across the retained view classifiers and selecting the class with the largest averaged probability. The retained set can include all trained views or a validation-selected subset when greedy model selection is enabled.

This inference-time aggregation combines complementary fragment-level rationales while preserving atom-level molecular prediction.

IV. Experiments

A. Research Questions

As our project investigates the effectiveness, design choices, and interpretability of fragment-based invariant subgraph learning for molecular out-of-distribution (OOD) prediction, we organize our experiments around four research questions.

- RQ1: OOD Effectiveness. How does fragment-based invariant subgraph learning affect OOD molecular-property prediction relative to the compared molecular graph baselines and fragmentation configurations?
- RQ2: Fragmentation and Multi-view Design. How do Group Graph, BRICS, RECAP, and mixed multi-view configurations affect OOD predictive performance?
- RQ3: Rationale Interpretability. Do selected fragment rationales identify stable, chemically meaningful molecular motifs across runs, and how do those motifs differ by fragmentation strategy and molecular label?
- RQ4: Sensitivity and Ablation. Which modeling choices most affect predictive performance and the size, agreement, or stability of selected rationales?

B. Dataset

We evaluate our method on three DrugOOD datasets: *DrugOOD-lbap-core-ec50-assay, DrugOOD-lbap-core-ec50-scaffold*, and *DrugOOD-lbap-core-ec50-size*. The datasets provide predefined training, validation, and test splits designed to measure generalization under complex distribution shifts, which in these datasets are across different assays, scaffolds, and molecule sizes. The data was directly taken from the curator provided by the original authors [6].

Each molecule retains the domain identifier provided by DrugOOD, which represents the environment information associated with each sample. Domain identifiers are used to define environmental groups for invariant learning objectives, specifically for IRM loss.

C. Compared Fragmentation & View Configurations

Each fragmentation strategy creates a separate fragment graph over the same underlying atom graph while preserving the original molecular structure for downstream reconstruction. The evaluated fragmentation strategies include Group Graph, BRICS, and RECAP. We additionally evaluate mixed configurations that combine multiple fragmentation strategies to investigate whether different fragment representations provide complementary predictive evidence.

For multi-view experiments, each fragmentation strategy is assigned multiple independent views. Views using the same fragmentation method operate on aligned fragment representations and aggregate their evidence, while different fragmentation strategies maintain separate fragment spaces and combine predictions through soft voting. The number of views and allocation between fragmentation strategies were split evenly for evaluation.

D. Hyperparameter Tuning & Evaluation

Hyperparameters were selected through an Optuna-based categorical search. The search evaluated 100 Optuna trials for 40 epochs each. The objective function maximized OOD validation ROC-AUC, with the best configuration selected based on validation performance. Each trial was repeated twice during hyperparameter search, and the averaged validation ROC-AUC was used for parameter selection.

For each Optuna trial, model checkpoints were evaluated using a moving-average validation criterion with a window size of five epochs to reduce sensitivity to individual validation fluctuations. After selecting the best hyperparameter configuration, the final model was evaluated five times over 100 epochs using fixed random seeds. Final test results are reported as the mean and standard deviation across these repeated evaluations.

The search space included both architectural and optimization parameters:

- $\mathcal{L}_{con} \in \{0.0, 0.1, 0.5, 1.0\}$
- $\mathcal{L}_{div} \in \{0.0, 0.5, 1.0, 2.0, 4.0\}$
- $\mathcal{L}_{IRM} \in \{0.0, 1.0, 2.0, 4.0, 8.0\}$
- IRM environments $\in \{2, 4, 8\}$
- Edge ratio $\in \{0.3, 0.4, 0.5, 0.6\}$
- Dropout $\in \{0.1, 0.2, 0.3, 0.4\}$
- GNN layers $\in \{2, 3\}$
- Hidden dimension $\in \{64, 128, 256\}$
- Batch size $\in \{64, 128, 256\}$
- Weight decay $\in \{10^{-4}, 10^{-3}, 10^{-2}\}$

E. Interpretability

To evaluate the interpretability of the learned fragment-level rationales, we analyze the molecular subgraphs selected during inference. For each molecule, the selected fragment edges are reconstructed into atom-level rationales using the stored fragment-to-atom mappings, allowing the model-selected evidence to be visualized in the original molecular representation.

We compare reconstructed rationales from our fragment-based approach with atom-level rationales extracted by baseline methods to examine whether fragment selection produces more chemically structured subgraphs. In particular, we investigate whether selecting fragment relationships preserves recognizable molecular structures rather than dividing individual chemical motifs across selected and unselected edges.Beyond qualitative visualization, we analyze the frequency of selected fragments and connected fragment pairs across molecules. Selection rates are grouped by model view, fragmentation strategy, and molecular label to identify recurring structures associated with different prediction outcomes.

V. Results & Discussion

A. OOD Predictive Performance

TABLE I
OOD PREDICTIVE PERFORMANCE MEASURED BY ROC-AUC. RESULTS ARE SUMMARIZED WITH MEAN ± STANDARD DEVIATION.

Datasets	EC-Assay	EC-Scaffold	EC-Size
SuGAr	78.62±0.84	65.56±0.43	66.36±1.06
Our Model	68.42±1.84	67.38±0.92	63.74±0.66

The performance of our model relative to the baseline model SuGAr is presented in table 1. Results for our model were obtained across 5 runs on set seeds, consistent with SuGAr's methodology. Our model achieves higher performance on the EC-Scaffold dataset, while SuGAr outperforms on the EC-Assay and EC-Size datasets.

The disparity in performance across the evaluation sets suggests that our approach is highly sensitive to the nature of the distribution shift. Future investigation will look into more detailed metrics of the datasets, such as fragment graph structure and class weight imbalance, and the impact of these properties on model performance.

B. Effect of Fragmentation & Multi-view Design

We did not get to work on this and thoroughly test different fragmentation techniques on different varying number of views. We would share results on OOD predictive performance based on the number of views used and the fragmentation method used on those views, comparing with baselines and reported results. Preliminary informal observations can be found in the *Ablation & Sensitivity Results* section.

C. Rationale Interpretability Results

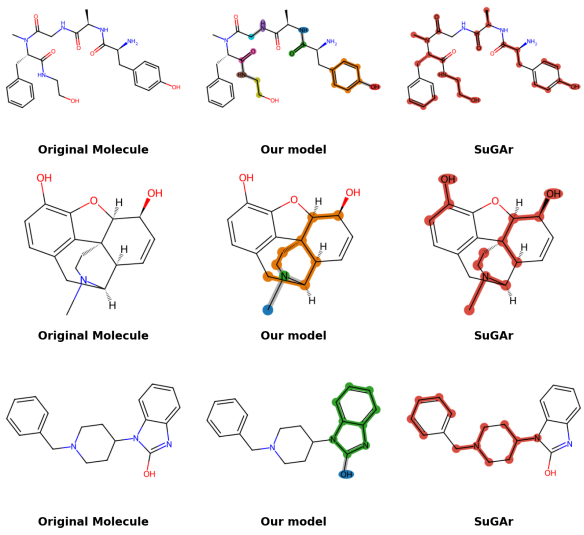


Fig. 3. **Inference rationale comparison.** Left: original molecule. Center: the atom-level rationale reconstructed from selected Group Graph fragments during inference. Right: the atom-level rationale selected by the SuGAr pipeline during inference.

To evaluate the interpretability of our model, we analyze the subgraphs extracted during inference. Figure 3 shows a comparison of the edges selected by our model and SuGAr on various molecules. The figure demonstrates that our model successfully selects for chemically meaningful substructures to drive the predictive decisions, while SuGAr's extracted patterns tend to break chemical motifs (most visible in the ring structures) and are therefore more difficult to interpret intuitively.

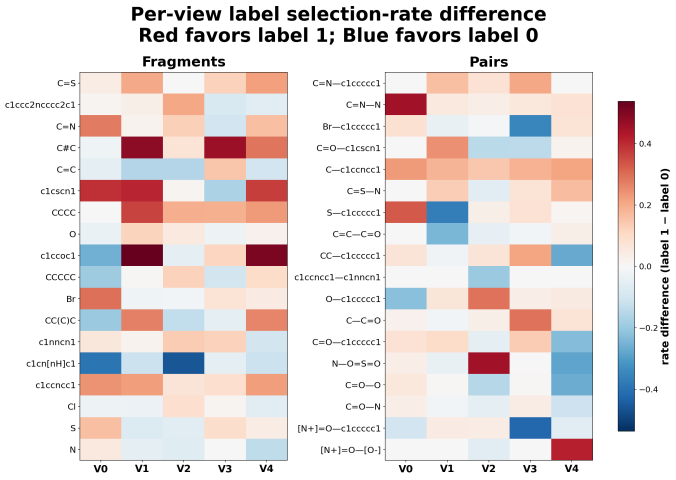


Fig. 4. **Fragment (left) and fragment-pair (right) selection rates across different views**. Darker colors indicate a stronger association with a particular label.

Figure 4 illustrates the selection rates of the most selected-for individual fragments and fragment edges across different views. Certain fragments and fragment-pair combinations consistently exhibit high selection probabilities tied to specific class labels, indicating that the fragment-CIGA objective successfully isolates recurring predictive motifs. These results not only verify that the model learns chemically sound structures but also provide domain experts with actionable hypotheses for targeted molecular optimization and causal validation.

D. Ablation & Sensitivity Results

While we were not able to conduct extensive ablation studies, which will be considered for future investigation, preliminary studies were done on loss components, view count, edge sampling ratio, and fragmentation strategies.

TABLE II
ABLATION STUDIES ON LOSS COMPONENTS, MEASURED BY ROC-AUC.

Performance	**IID**	**OOD**
Baseline (no losses)	81.29	70.14
Diversity only	80.33	73.28
Contrastive only	78.07	70.46
IRM only	81.92	74.86

Table II shows the results of single runs for ablation testing on loss components. Results suggest significant above-baseline OOD performance for diversity loss and IRM loss, but not contrastive loss. However, as these tests were conducted with only one run each, they may be subject to significant noise, so further studies with repeated trials are needed for concrete performance assessment.

Other informal preliminary testing suggested that:

- Performance was not impacted significantly for view counts between 5-10, but decreased significantly when lowered to 3.
- Increasing the edge sampling ratio from around 0.05-0.15 to 0.3 did not impact test results, but increased training and validation performance.
- BRICS-only, RECAP-only, and mixed-fragmentation runs resulted in overall worse performance than Group Graph-only.

VI. CONCLUSION

In this project, we explored a fragment-based subgraph learning approach for OOD generalization and chemical interpretability in molecular graph prediction. By utilizing fragmentation with the Group Graph method, we successfully produce subgraphs that retain meaningful chemical structure, yielding more interpretable predictive rationales compared to baseline models like SuGAr. In addition, our selection-rate analysis identifies label-associated model evidence that can guide chemical interpretation and future causal validation.

Although our model achieved increased performance compared to SuGAr on one dataset, it struggled to perform competitively on other evaluated datasets. Future work will further examine hyperparameter optimization, extracted chemical motifs, training features, and the overall pipeline to improve overall molecular OOD generalization.

VII. CONTRIBUTIONS

A. Lance Santiago

- Molecular fragmentation and preprocessing foundation
- Dataset parsing
- Group Graph pipeline
- BRICS and RECAP integration
- Atom and fragment provenance and mappings fields
- Implemented mean pooling during preprocessing for initial feature vectors
- Support for DrugOOD datasets
- Diagnostics on fragments
- Fragment edge sampler implementation
- Implemented atom-graph reconstruction
- Consolidated configuration file
- Expanded CLI support (argparse stuff)
- Checkpoints
- Mixed fragmentation support
- Final output metrics
- Fragmentation selection metrics (label aware)
- Loss tracking (including diagnostics and plotting)
- Poster: Wrote intro, background, experimental design and conclusion.
- Made general outline for this report/paper

- Wrote initial abstract
- Brought donuts for everyone.

B. Nancy Yang

- SuGAr pipeline (training pipeline + evaluation pipeline)
- Model separation into subgraph extractor (FragmentCIGA), reconstruction, and classifier (AtomClassifier)
- Subgraph aggregation and soft voting
- Diversity, IRM, contrastive, and prediction losses
- Diversity loss fix to exclude dropped edges
- Greedy model selection
- ROC-AUC tracking
- Best performing epoch selection
- Training speed-up through vectorizing functions
- IRM environment splits based on domain IDs
- Visualization of extracted subgraphs
- Hyperparameter search script and config file
- Evaluation and results averaging script
- Moving average tracking for early stopping
- Analysis of the top extracted fragments + edges
- Heat map visual of fragment selection rate per view
- Visualization of our subgraphs vs. SuGAr's
- Ran model often to debug losses and hyperparameters
- Final results
- Poster: layout, figures, and tables
- Model pipeline visual for poster + model description

C. Shared Contribution

- Integrated preprocessing with SuGAr pipeline
- Poster: general cleanup, fix phrasing

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